

Report Part Title: Big Box Retail Decentralization in the Western United States

Report Title: The Influence of Local Fiscal Structure and Growth Control Choices on
“Big Box” Urban Sprawl in the American West

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Big Box Retail Decentralization in the Western United States

An empirical study of the degree of metropolitan retail decentralization must begin with a unit of analysis. For this study it is the 61 metropolitan areas in what the Census Bureau defined in 1990 as the continental western United States, less the seven metropolitan areas in Idaho, Montana, and Wyoming. The analysis is limited to metropolitan areas in the West for a few reasons. The first is that six of the eight states defined as Western enacted statewide ballot box or legislative restrictions on the local use of property taxes between 1977 and 1997. Through Proposition 13 (1978), Measures 5 and 47 (1990 and 1996), and Amendment 1 (1992); California, Oregon, and Colorado voters all used the citizen initiative to limit local property taxation. In Arizona, Nevada, and Utah, state legislatures took similar steps.⁴ These restrictions, which Sokolow (2000) classifies as harsher than in any other region in the United States, offers natural experiments by which to test the influence of changes in statewide municipal fiscal structure on metropolitan retail decentralization. Furthermore, most metropolitan areas in the western United States grew up in an era of rising populations, rising real incomes, and declining transportation costs. Metropolitan areas in Idaho, Montana, and Wyoming are not included because the metropolitan development patterns in these three states are very different in comparison to the western states retained in the sample used here.⁵

Fifty-four metropolitan areas are used in the analysis. A metropolitan area consists of the relevant component counties in the 1990 Census definition of either a Metropolitan Statistical Area (MSA) or a Primary Metropolitan Statistical Area (PMSA).⁶ Since the focus of this paper is retail activity in suburban locations, the suburban area within a metropolitan area is the constituent counties in a MSA or PMSA, less the “central places” included in the 1990 definition of urbanized areas in a metropolitan area.⁷ Central places are considered by the Census to be the dominant employment and residential centers in each urbanized area. For instance, the suburban area in the San Diego MSA would be San Diego County less the cities of Coronado, Escondido, and San Diego. Table 1 contains a list of the 54 metropolitan areas in the sample, the areas’ component counties, and the central places that are excluded from these counties to create the definition of a metropolitan area’s non-central places that is used here. This designation of suburbia is

⁴ Sokolow (2000) offers a comprehensive survey of property tax limitation in the western United States. See Chapman (1998) for a summary of the local public finance consequences of California’s 1978 passage of Proposition 13.

⁵ The largest central cities in each of these excluded states only had 1992 populations of 136,000, 84,000, and 52,000 respectively.

⁶ A PMSA consists of integrated counties that are divisible into smaller integrated units that consist of one or more counties. A MSA consists of counties that are not divisible into smaller integrated units.

⁷ This approach should be considered conservative in regard to defining excessive decentralization because many would consider the location of retail activity in a place not a central place in 1977 and 1987, but classified as such in 1990, as sprawl.

an attempt to account for the polycentric nature of most U.S. metropolitan areas through the use of existing data sources.

The three dependent variables in the regression analyses that follow are (1) total retail sales, (2) building material sales, and (3) new and used auto sales. Big box retail sales are represented by building material and new/used auto sales due to popular perceptions of what big box is (“Home Depots” and “Automalls”) and the availability of comparable data for 1977, 1987, and 1997 for the 54 metropolitan areas under consideration.⁸ Table 2 offers a comparison of the ratio of non-central place building material (auto) sales to total metropolitan area building material (auto) sales for the 47 metropolitan areas in the sample for which data is available.⁹ The top eight data rows in this table report the averages for each state, where available, using metropolitan area as the unit of observation.

As shown at the top of Table 2, for building material and auto sales where multiple year averages are available, some western states (5 out of 12 cases) experienced an increase in the percentage of these forms of big box sales occurring in non-central locations relative to all metropolitan locations, while the other western states (7 out of 12 cases) experienced a decrease. Just based on these percentages it is hard to distinguish whether these forms of urban retail sprawl have increased or decreased over time. But as indicated above, it is perhaps best to not designate an increase in the percentage of retail activity occurring in non-central places as sprawl if population, income, and demographic shifts warrant such changes. In addition, there are many distinct metropolitan areas that experienced a decrease in the percentage of building material sales occurring in non-central places between 1987 and 1997 (Chico-Paradise, Sacramento, Denver, Bellingham, and Tacoma) and even more for auto sales (Chico-Paradise, Denver, Las Vegas, Medford-Ashland, Salem, and Spokane). Other than that there is a great deal of variation in the degree of non-central place big box retail activity occurring in western metropolitan areas and over time, it is hard to draw any specific conclusions from the information in Table 2. A regression analysis of the determinants of suburban retail activity is necessary to comprehend the causes of the observed variation.

⁸ Data gathering was complicated by the fact that the U.S. Census Bureau switched from using Standard Industrial Codes (SIC) in 1987 to designate different types of retail activity, to North American Industrial Classification System (NAICS) codes in 1997. For many types of retail this made switch made multi-year comparisons impossible. This was not a problem for new and used auto sales that is represented by SIC 551 (motor vehicle dealer—new and used cars) in 1977 and 1987, and NAICS 4411 (automobile dealers—new and used) in 1997. Building materials is represented by SIC 521,3 (building materials and supply stores) and SIC 525 (hardware stores) in 1977 and 1987, and NAICS 4441 (building materials and supply dealers) in 1997.

⁹ Table I in Wassmer (forthcoming) offers similar information for aggregate retail sales for all 54 of the western state metropolitan areas in the full sample.

Table 1: Urban Area Definitions for the Western United States

1990 Metropolitan Area Name	1990 Counties in Metropolitan Area	1990 Central Places (Cities) in Metropolitan Area	1990 Metropolitan Area Name	1990 Counties in Metropolitan Area	1990 Central Places (Cities) in Metropolitan Area
Phoenix-Mesa AZ, MSA	Maricopa AZ, Pinal AZ	Mesa AZ, Phoenix AZ, Scottsdale AZ, Tempe AZ	San Diego CA, MSA	San Diego CA	Coronado CA, Escondido CA, San Diego CA
Tucson AZ, MSA	Pima AZ	Tucson AZ	San Francisco CA, PMSA	Marin CA, San Francisco CA, San Mateo CA	San Francisco CA
Yuma AZ, MSA	Yuma AZ	Yuma AZ	San Jose CA, PMSA	Santa Clara CA	Gilroy CA, Palo Alto CA, San Jose CA, Santa Clara CA, Sunnyvale CA
Bakersfield CA, MSA	Kern CA	Bakersfield CA	San Luis Obispo-Atascadero-Paso Robles CA, MSA	San Luis Obispo CA	Atascadero CA, Paso Robles CA, San Luis Obispo CA
Chico-Paradise CA, MSA	Butte CA	Chico CA	Santa Barbara-Santa Maria-Lompoc CA, MSA	Santa Barbara CA	Lompoc CA, Santa Barbara CA, Santa Maria CA
Fresno CA, MSA	Fresno CA, Madera CA	Fresno CA	Santa Cruz-Watsonville CA, PMSA	Santa Cruz CA	Santa Cruz CA, Watsonville CA
Los Angeles-Long Beach CA, PMSA	Los Angeles CA	Lancaster CA, Long Beach CA, Los Angeles CA, Pasadena CA	Santa Rosa CA, PMSA	Sonoma CA	Petaluma CA, Santa Rosa CA
Merced CA, MSA	Merced CA	Merced CA	Stockton-Lodi CA, MSA	San Joaquin CA	Lodi CA, Stockton CA
Modesta CA, MSA	Stanislaus CA	Modesto CA, Turlock CA	Vallejo-Fairfield-Napa CA, PMSA	Napa CA, Solano CA	Fairfield CA, Napa CA, Vacaville CA, Vallejo CA
Oakland CA, PMSA	Alameda CA, Contra Costa CA	Alameda CA, Berkeley CA, Oakland CA	Ventura CA, PMSA	Ventura CA	San Buenaventura (Ventura) CA
Orange CA, PMSA	Orange CA	Anaheim CA, Irvine CA, Santa Ana CA	Visalia-Tulare-Porterville CA, MSA	Tulare CA	Porterville CA, Tulare CA
Redding CA, MSA	Shasta CA	Redding CA	Yolo CA, PMSA	Yolo CA	Davis CA, Woodland CA
Riverside-San Bernardino CA, PMSA	Riverside CA, San Bernardino CA	Hemet CA, Palm Dessert CA, Palm Springs CA, Riverside CA, San Bernardino CA, Temecula CA	Yuba City CA, MSA	Sutter CA, Yuba CA	Yuba CA
Sacramento CA, PMSA	El Dorado CA, Placer CA, Sacramento CA	Sacramento CA	Boulder-Longmount CO, PMSA	Boulder CO	Boulder CO, Longmount CO

1990 Metropolitan Area Name	1990 Counties in Metropolitan Area	1990 Central Places (Cities) in Metropolitan Area	1990 Metropolitan Area Name	1990 Counties in Metropolitan Area	1990 Central Places (Cities) in Metropolitan Area
Salinas CA, MSA	Monterey CA	Monterey CA, Salinas CA	Colorado Springs CO, MSA	El Paso CO	Colorado Springs, CO
Denver CO, PMSA	Adams CO, Arapahoe CO, Denver CO, Douglas CO, Jefferson CO	Denver CO	Salt Lake City-Ogden UT, MSA	Davis UT, Salt Lake UT, Weber UT	Salt Lake City UT, Ogden UT
Fort-Collins-Loveland CO, MSA	Larimer CO	Fort Collins CO	Bellingham WA, MSA	Whatcom WA	Bellingham WA
Grand Junction CO, MSA	Mesa CO	Grand Junction CO	Bremerton WA, PMSA	Kitsap WA	Bremerton WA
Greeley CO, MSA	Weld CO	Greeley CO	Olympia WA, PMSA	Thurston WA	Olympia WA
Pueblo CO, MSA	Pueblo CO	Pueblo CO	Richland-Kennewick-Pasco WA, MSA	Benton WA, Franklin WA	Kennewick WA, Pasco WA, Richland WA
Las Vegas NV & AZ, MSA	Clark NV, Mohave AZ, Nye NV	Las Vegas NV	Seattle-Bellevue-Everett WA, PMSA	Island WA, King WA, Snohomish WA	Auburn WA, Everett WA, Seattle WA
Reno NV, MSA	Washoe NV	Reno NV	Spokane WA, MSA	Spokane WA	Spokane WA
Albuquerque NM, MSA	Bernalillo NM, Sandoval NM, Valencia NM	Albuquerque NM	Tacoma WA, PMSA	Pierce WA	Tacoma WA
Las Cruces NM, MSA	Dona Ana NM	Las Cruces NM	Yakima WA, MSA	Yakima WA	Yakima WA
Santa Fe NM, MSA	Los Alamos NM, Santa Fe NM	Santa Fe NM			
Eugene-Springfield OR, MSA	Lane OR	Eugene OR, Springfield OR			
Medford-Ashland OR, MSA	Jackson OR	Medford OR			
Portland-Vancouver OR, PMSA	Clackamas OR, Columbia OR, Multnomah OR, Washington OR, Yamhill OR, Clark WA	Portland OR, Vancouver WA			
Salem OR, PMSA	Marion OR, Polk OR	Salem OR			
Provo-Orem UT, MSA	Utah UT	Provo UT, Orem UT			

Table 2: Distribution of Building Material and Auto Sales for Non-Central Places in Metropolitan Areas in the Western United States

1990 Metropolitan Area Name	1977 Non-Central Places Building Mat. Sales / Metro Building Mat. Sales	1987 Non-Central Places Building Mat. Sales / Metro Building Mat. Sales	1997 Non-Central Places Building Mat. Sales / Metro Building Mat. Sales		1977 Non-Central Places Auto Sales / Metro Auto Sales	1987 Non-Central Places Auto Sales / Metro Auto Sales	1997 Non-Central Places Auto Sales / Metro Auto Sales
Arizona average for MSAs			0.23			0.10	0.15
California average for (P)MSAs	0.81	0.59	0.51		0.50	0.53	0.45
Colorado average for (P)MSAs		0.40	0.38		0.20	0.35	0.33
Nevada average for MSAs		0.46	0.51			0.57	0.25
New Mexico average for MSAs		0.26	0.04				0.13
Oregon average for (P)MSAs			0.33		0.37	0.25	0.27
Utah average for MSAs			0.10		0.46	0.73	0.53
Washington average for (P)MSAs	0.65	0.55	0.48		0.35	0.34	0.38
Phoenix-Mesa AZ, MSA			0.265			0.201	0.216
Tucson AZ, MSA			0.189			0.000	0.081
Bakersfield CA, MSA			0.592		0.239	0.195	0.296
Chico-Paradise CA, MSA		0.538	0.258			0.473	0.406
Fresno CA, MSA			0.308			0.409	0.467
LA-Long Beach CA, PMSA			0.629			0.618	0.699
Orange CA, PMSA			0.670			0.708	
Ventura CA, PMSA	0.913				0.677	0.701	0.789
Merced CA, MSA			0.452			0.176	
Modesta CA, MSA							0.484
Redding CA, MSA			0.070				
Sacramento CA, PMSA		0.803	0.704			0.781	0.877

1990 Metropolitan Area Name	1977 Non-Central Places Building Mat. Sales / Metro Building Mat. Sales	1987 Non-Central Places Building Mat. Sales / Metro Building Mat. Sales	1997 Non-Central Places Building Mat. Sales / Metro Building Mat. Sales		1977 Non-Central Places Auto Sales / Metro Auto Sales	1987 Non-Central Places Auto Sales / Metro Auto Sales	1997 Non-Central Places Auto Sales / Metro Auto Sales
Salinas CA, MSA		0.422			0.416		
San Diego CA, MSA							0.474
Oakland CA, PMSA						0.782	
San Francisco CA, PMSA	0.706	0.606	0.677		0.657	0.737	0.814
San Jose CA, PMSA							0.098
Santa Rosa CA, PMSA			0.497				0.160
Vallejo-Fairfield-Napa CA, PMSA							0.080
Visalia-Tulare-Porterville CA, MSA			0.782				
SLO-Atasc-Paso Robles CA, MSA			0.669				
San Barb-Santa Maria-Lom CA, MSA							0.221
Stockton-Lodi CA, MSA			0.295			0.272	0.383
Boulder-Longmount CO, PMSA							0.153
Colorado Springs CO, MSA		0.080	0.118				0.003
Denver CO, PMSA		0.712	0.597			0.875	0.806
Fort Collins-Loveland CO, MSA			0.427		0.388	0.327	0.391
Grand Junction CO, MSA						0.194	
Greeley CO, MSA					0.219		0.301
Pueblo CO, MSA					0.000	0.000	
Las Vegas NV & AZ, MSA			0.548			0.567	0.490
Reno NV, MSA		0.455	0.469				0.013
Albuquerque NM, MSA		0.256					
Las Cruces NM, MSA			0.037				
Santa Fe NM, MSA							0.130

1990 Metropolitan Area Name	1977 Non-Central Places Building Mat. Sales / Metro Building Mat. Sales	1987 Non-Central Places Building Mat. Sales / Metro Building Mat. Sales	1997 Non-Central Places Building Mat. Sales / Metro Building Mat. Sales		1977 Non-Central Places Auto Sales / Metro Auto Sales	1987 Non-Central Places Auto Sales / Metro Auto Sales	1997 Non-Central Places Auto Sales / Metro Auto Sales
Eugene-Springfield OR, MSA							0.340
Medford-Ashland OR, MSA			0.334			0.222	0.202
Salem OR, PMSA			0.330		0.368	0.287	0.263
Provo-Orem UT, MSA			0.095				0.295
Salt Lake City-Ogden UT, MSA					0.456	0.730	0.773
Bellingham WA, MSA		0.417	0.252			0.255	
Bremerton WA, PMSA			0.556				0.304
Olympia WA, PMSA			0.502				0.060
Seattle-Bellevue-Everett WA, PMSA			0.668				0.671
Spokane WA, MSA	0.551		0.388		0.333	0.366	0.341
Tacoma WA, PMSA	0.743	0.681	0.516		0.395	0.470	0.578
Yakima WA, MSA					0.326	0.276	0.318